Mittuniversitetet
MID SWEDEN UNIVERSITY

# Försättsblad Prov Original

| Kurskod | Provkod | Tentamensdatum |
|---|---|---|
| MA080G | T102 | 2026-01-05 |
| Kursnamn | Matematik GR (B), Kryptografi | |
| Provnamn | Skriftlig tentamen | |
| Ort | Sundsvall | |
| Termin | | |
| Ämne | | |

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# Exam – Cryptography (MA080G)

Mittuniversitetet
Department: IMD
Responsible: Aurélio Menegon

Date: January 5, 2026
Time: 5 hours
No books, notes or symbolic calculators allowed.
Calculator make/model must be stated on the cover page.

## Instructions

Answer each subquestion on a separate line. Complete solutions must be provided – reasoning and calculations must be clear and not overly brief. **ALL ANSWERS MUST BE JUSTIFIED.** Grades are based on the achievement of the course objectives.

**Grading scale:** A (22p), B (18p), C (14p), D (10p), E (9p)

## Problems

### 1. Classical Ciphers (4p)

(a) (2p) A Caesar cipher maps the letter "E" to "H". What is the encryption key? Then, use it to encrypt the word "DATA".

(b) (2p) Decrypt the ciphertext "RIJVSUYVJN" using the Vigenère key "KEY".

### 3. Modular Arithmetic and Groups (8p)

(a) (1p) Compute $\phi(105)$ using its prime factorization.

(b) (1p) Determine whether 4 is a unit in $\mathbb{Z}/35\mathbb{Z}$. Justify.

(c) (1p) List all generators of $(\mathbb{Z}/12\mathbb{Z}, +)$.

(d) (1p) Find the order of 7 in $(\mathbb{Z}/15\mathbb{Z})^{\times}$.

(e) (1p) Let $a = 11$ and $n = 40$. Use Euler's Theorem to compute $a^{\phi(40)} \bmod 40$.

(f) (2p) Compute the modular inverse of 17 modulo 43. Show all steps.

(g) (1p) Solve the congruence $14x \equiv 12 \mod 22$ or explain why no solution exists.

## 4. Diffie–Hellman Key Exchange (3p)

Let $p = 31$, $g = 3$, Alice's private key is $a = 7$, and Bob's private key is $b = 12$.

(a) (1p) Compute Alice's public key.

(b) (1p) Compute Bob's public key.

(c) (1p) Compute the shared secret.

## 5. ElGamal Encryption (3p)

Let $p = 103$, $g = 5$, and suppose Alice's private key is $a = 21$. A message is encrypted using her public key and sent as a ciphertext value $c = 34$, where the encryption follows the simplified form $c = m \cdot h \mod p$, and $h = g^a \mod p$.

(a) (1p) Compute Alice's public key $h$.

(b) (1p) Compute the modular inverse of $h \mod p$.

(c) (1p) Use your result to decrypt the message $m = c \cdot h^{-1} \mod p$.

## 6. RSA Encryption and Decryption (4p)

Let $n = 221$, $e = 5$.

(a) (1p) Factor $n$ and compute $\phi(n)$.

(b) (1p) Find the private exponent $d$ such that $ed \equiv 1 \mod \phi(n)$.

(c) (1p) Encrypt the message $m = 13$.

(d) (1p) Decrypt the ciphertext $c = 91$.

## 7. Algorithm for Factoring (2p)

Use Pollard's $\rho$ algorithm to find a nontrivial factor of $n = 221$, using the iteration function $f(x) = x^2 + 1 \bmod 221$ and starting value $x_0 = 2$. Perform the necessary steps of the algorithm, showing the values of $x_i$, $y_i = f(f(x_{i-1}))$, and $\gcd(|x_i - y_i|, 221)$ at each step.